Building Knowledge Graphs: From Sources to Verifiable Queries

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1 Introduction

A knowledge graph connects identified things and the relationships claimed between them. Its value is not that drawing an edge makes a statement true. Its value is that a reader can follow a chain from a question through explicit modelling decisions to the source records on which an answer depends. This primer develops that chain: from source material and uncertain names to qualified claims, evidence-backed queries, validation, and repeatable releases.

The reader is a software or data practitioner comfortable with tables and basic data modelling. No prior graph-database or ontology experience is assumed. The main route follows Northbridge, a synthetic organisational-policy case, from source evidence to a bounded set of claims and then to questions whose answers can be inspected. This is a guide to the reasoning behind a knowledge graph, not a complete reference for database products, formal logic, or legal interpretation. Policy authority in the example is an explicit modelling assumption, not legal advice.

Read the primer as a standalone explanation. The central case is worked with tables, small graph diagrams, and prose before any implementation detail is introduced. The Python, SQL, R, RDF, and property-graph material later in the primer is companion evidence for particular representations; you should be able

to understand the model, reproduce the reasoning by hand, and judge the answer limits without opening a repository file.

The route is deliberately divided into two questions. First, how do we build a graph without turning a source statement into an unsupported fact? Second, how do we use the resulting graph without confusing a path, a stored assertion, and a conclusion? A final section covers validation, evaluation, correction, release, privacy, and responsibility. RDF, property graphs, and R are useful comparison branches, not prerequisites for the main case.

Verified against primary sources for Construct and Query with Python and DuckDB; Represent Claims in RDF and Property Graphs; Analyse a Projection in R; and Validate, Evaluate, and Diagnose; unverified claims are flagged inline.

2 What a Knowledge Graph Is—and When It Helps

A knowledge graph is a model of entities and relationships in which the relationships can be inspected, traversed, and qualified. That definition is intentionally modest. A graph does not become knowledge merely because its data is drawn as circles and arrows, and it does not become trustworthy merely because a graph database stores it. The useful question is: what can a reader ask, which evidence supports the answer, and which modelling decisions stand between the source and the result?

Consider three ways to record the same small fact:

Representation	What it makes easy to see	What remains unresolved
Sentence: “The register says version 1 applies to Atlas.”	The wording and its source context	Whether “applies” has a date, what version of the register was read, and how to compare a conflicting sentence

Representation	What it makes easy to see	What remains unresolved
Row: `(policy:travel-v1, applies_to, project:atlas)`	The endpoints and predicate	Who asserted it, when it applies, whether it is positive or negative, and what the reader should do when another source disagrees
Qualified graph claim	Endpoints, predicate, source, locator, interval, polarity, and decision status can be queried together	The graph still depends on the quality and coverage of its source boundary and modelling rules

The third representation is the focus here. It is not always the best storage choice. A small, stable report may be better kept as a document; a fixed set of joins may be clearer in ordinary relational tables; a graph becomes attractive when the important questions follow relationships, combine evidence from different sources, or need to retain uncertainty about those relationships.

2.1 The smallest useful graph

Start with a graph that answers one real question rather than with a list of technologies. Suppose a directory records that Atlas belongs to Northbridge and a policy register records that Travel Policy v1 applies to Atlas. The unqualified picture is:

```
Travel Policy v1 ──applies_to──► Atlas ──belongs_to──► Northbridge
```

This path answers a structural question: is the policy connected to the organisation through the project? It does not answer whether the policy was valid on a particular date, whether both edges came from the same source, or whether the path is an authorised inference. For those questions, each edge needs an evidence-bearing record. A useful hand-worked version is:

Claim	Subject	Predicate	Object	Source	Scope	Polarity
c1	policy:travel-v1	applies_to	project:atlas	policy register, R1:P1-P2	1 January–1 April 2024, end exclusive	positive
c3	project:atlas	belongs_to	org:northbridge	directory, D1:A1	source record's stated scope	positive

The path can now be returned together with the two claim IDs and their locators. That is the first important difference between a graph used for evidence-backed work and a graph used only as a visualisation: the answer can be challenged one relationship at a time.

2.2 Why the graph is more than a set of edges

The graph's entities and relationships are only the visible centre of a larger model. A source contains *mentions*: occurrences such as "Alex Kim" or "version 2". An entity is the thing the model may identify from that mention. A claim says that a qualified relationship is being considered; an assertion says that a particular source or process presents it. These are not synonyms. Keeping them separate lets the graph represent both "the minutes deny that version 1 applies" and "the register says that it does" without overwriting one source with the other.

Qualification carries the context needed to interpret the relationship. Common qualifiers include:

- source and locator, so a reviewer can return to the evidence;
- valid time and observation or publication time, which answer different temporal questions;
- polarity, so a denial is not silently discarded;
- confidence or review state, when those are defined and calibrated;
- audience, access, or jurisdiction, when a relationship is not universal;

- extraction and identity decisions, so a correction can be traced to its intermediate reasoning.

Not every graph needs every field. The discipline is to begin with the questions and retain the fields that those questions require. Adding a generic "confidence" number does not solve an identity ambiguity if the source never provides a distinguishing identifier. Adding a graph database does not solve a missing date. The model should make those limits visible rather than decorate them with infrastructure.

2.3 When a graph helps—and when it does not

A graph is useful when relationships are central to the reader's task. Typical questions include:

1. Which things are connected through a particular relation, and what are the intermediate steps?
2. Which sources support each step of a path?
3. Do competing sources make incompatible claims about the same relationship?
4. What was supported at a stated date, and what is merely unknown afterwards?
5. Which related records must be reconsidered when a source or identity decision changes?

These are not automatically graph-database questions. A relational design can answer them with tables and joins, especially when the depth and predicates are known in advance. The graph vocabulary is valuable because it makes the relationship structure and path semantics explicit. Choose a serving engine only after the questions, evidence fields, update pattern, and access boundary are clear.

The Northbridge case keeps this distinction visible. It uses a tabular ledger for the canonical evidence-preserving route because the teaching problem is to show the records and decisions clearly. A later comparison explains how the same qualified claims could be mapped to RDF or a property graph. The reader does not need to believe that one representation is universally superior; the reader needs to understand what information each representation retains or loses for a particular question.

The rest of the primer therefore follows a simple promise: define the question and evidence boundary, build claims conservatively, use those claims to answer bounded queries, and state what the answer cannot establish. Every example is small enough to inspect by hand before it is connected to an implementation.

3 Define the Boundary Before Building the Graph

A knowledge graph is an answer-producing model over a declared evidence set. Its boundary therefore includes more than the files that happened to be ingested. It includes the questions the model is intended to answer, the source classes it accepts, the period and population those sources cover, and the decisions it is authorised to make when records disagree. If any of these are implicit, users can mistake a bounded extract for a complete account.

3.1 Source Roles and Competency Questions

Northbridge makes the boundary concrete. The fixture contains an authorised policy register, a Northbridge directory, and Atlas meeting minutes. It represents two policy versions, two projects, Morgan Lee, and two distinct people named Alex Kim. For the teaching scenario, the register is treated as the designated source for recorded policy applicability and ownership; the directory supports project affiliation; the minutes represent what a meeting record states. This is a modelling assumption, not a source-ranking rule implemented by the script. Those roles do not make the register universally authoritative about every question. The minutes are the primary evidence for the sentence recorded at the meeting, even where the register is preferred to answer which version the organisation recorded as applicable.

Competency questions turn that boundary into observable requirements. For example: which policy version did the register say applied to Atlas on 1 March 2024? What owner did the register give for that version? What did the 15 March minutes say, and did that statement agree with the register for the same date? Can the attendee called Alex Kim be tied to either directory person? Each question names a relation or answer, and should also settle relevant scope: source authority, temporal interpretation, required evidence, and what to do when evidence is insufficient. A question that asks “Who is responsible?” leaves open whether responsibility means policy ownership, project sponsorship, or meet-

ing attendance.

Treat each question as a contract for a query and its result. The following contracts use only the supplied synthetic records. Every answer must carry its source scope and locator; an abstention must give its reason rather than silently omit a row. "Executable" means the stated decision is asserted by the Python example, while "illustrative" means the fixture contains the cited records but the full question is not asserted as a query.

Question and scope	Result contract on the supplied records	Status
Q1. Which version does the register record for Atlas on 1 March 2024?	Return version 1, `c1`, `s-register`, `R1:P1-P2`, the tested date and the register-only selection rule. This reports the register, not an organisation-wide governing decision.	Executable temporal classification for `c1`; the version-selection presentation is illustrative.
Q2. Who does the register name as owner of version 1?	Return Morgan Lee through `c2`, `s-register`, `R1:P1-P2`, with the owner relation's 1 January support horizon. Abstain on *continued* ownership as of 1 March: `c2` has an unknown end and supplies no later support.	Illustrative owner answer; `c2` is loaded but this question is not asserted by the executable.

Question and scope	Result contract on the supplied records	Status
Q3. What do the 15 March minutes state?	Return positive version 2 `c4` and negative version 1 `c5`, both from `s-minutes`, `M1:paragraph-3`, with their date, polarity and unknown end. Do not silently discard the negative sentence.	Source-derived rows executable; combined minutes answer illustrative.
Q4. Do the minutes and register agree for Atlas on 15 March?	Return `c1` positive against `c5` negative for version 1, both applicable on that date, with both locators and `abstain: conflicting evidence`. The teaching policy does not declare either source true.	Executable dated conflict result.
Q5. Which directory person was the attendee named Alex Kim?	Return the unresolved `m-alex` mention, `r2` abstention, both `person:alex-kim-1` and `person:alex-kim-2` candidates, and `M1:paragraph-3`; assign no person ID to attendance.	Executable bounded resolution decision; audience-specific disclosure is illustrative.

These contracts close the opening questions at their actual evidence limits. In particular, the recorded owner is an answer to a source question, while ownership on a later date is a different temporal question. A graph that always returns one Alex person has encoded an assumption, not demonstrated better coverage.

3.2 Work the source/date distinction for Q2

Q2 is deliberately different from Q1. The first question asks what the register records about applicability on a date inside `c1`'s bounded interval. The second asks what the register names as the owner and then tests whether that ownership can be carried forward to a later date. A reader should not answer both questions by selecting the same row and copying its date treatment.

Work the reasoning in four steps:

1. Select `c2` by the question's source scope (`s-register`), relation (`owned_by`), and object (`policy:travel-v1`). This gives Morgan Lee and the locator `R1:P1-P2`.
2. Read the temporal fields rather than treating the row as timeless. In the fixture, `c2` starts on 1 January and has an `unknown` end with support only through 1 January. The source supports the recorded owner on that date, not an unqualified assertion about every later date.
3. Apply the question date, 1 March, to the temporal contract. Because it is after `supported_through`, the result is `indeterminate`, not `supported` and not `refuted`.
4. Return both pieces of information: "the register names Morgan Lee on 1 January" and "continued ownership on 1 March is indeterminate from the supplied evidence". Include `c2` and its locator so a reviewer can inspect the same decision.

This is a useful manual query even though the bounded executable does not assert the complete owner presentation as an end-to-end operation. The distinction is the teaching point: an answer can be valid for a source record and still be insufficient for a later-date question. If a system returned Morgan Lee as the 1 March owner without the qualification, inspect temporal selection and answer policy rather than merging another person or changing the graph storage.

3.3 Completeness and Negative Answers

The boundary also governs negative answers. The absence of a matching claim means only that no claim was found in this fixture after applying the query's filters. It does not establish that Northbridge has no other policies, that no other meeting discussed the rule, or that Atlas has no other affiliations. To interpret absence as a negative fact, the source set needs a completeness guarantee for

a defined subject, predicate, period, and population. The small synthetic fixture offers no such guarantee. Its appropriate use is to demonstrate construction, conflict handling, and query behaviour on known records, not to estimate coverage of a real organisation.

This distinction prevents two common category errors. First, “the graph says” should not conceal which source and selection rule produced the result. Second, a missing path is not evidence that a relationship does not exist. State the fixture boundary alongside conclusions, and return “not represented in the supplied sources” when that is all the query establishes.

The teaching fixture is synthetic data, not a set of retained source documents. The table summarises its source records and selected assertions; the locators are exact source-level labels within those synthetic records. A production graph would need stable source versions and access to the retained evidence, not merely the labels shown here.

Source	Locator	Selected fixture content
Authorised policy register `s-register`	`R1:P1-P2`	`c1`: version 1 applies to Atlas from 1 January to 1 April 2024; `c2`: version 1 is owned by Morgan; `c6`: version 2 applies from 1 April.
Northbridge directory `s-directory`	`D1:A1`	`c3` and `c7`: Atlas and Orion belong to Northbridge. The shared source locator does not identify a separate Orion row.
Atlas meeting minutes `s-minutes`	`M1:paragraph-3`	From 15 March, `c4` asserts version 2 applies and `c5` denies that version 1 applies; `m-alex` is unresolved.

The teaching data stores the two Alex entities but not their directory identifiers

or complete register and minutes. The construction companion retains short synthetic excerpts and three directory rows for its executable slice. It records only version 1 ownership, not ownership of version 2. The start date on a meeting claim is an encoded valid-time interpretation, not a general rule that a publication date automatically fixes validity.

4 Represent Evidence, Entities, and Qualified Claims

The boundary tells us which source occurrences matter; it does not tell us which occurrences refer to the same thing. Before converting those occurrences into claims, keep their source evidence separate from the identities the graph may assign.

4.1 From source mentions to qualified claims

A mention is an occurrence in a source; an entity is a thing the model refers to; an identifier names that entity within a defined namespace. A label is a human-readable description and need not be unique. These objects have different lifecycles and should not be collapsed into one row just because a simple graph drawing has one node per thing.

In the fixture, `m-alex` records "Alex Kim" in `s-minutes` with a null entity. Two distinct entities, `person:alex-kim-1` and `person:alex-kim-2`, share that label. Resolution decision `r2` records both candidates and abstains because the meeting gives no distinguishing identifier. The occurrence survives without inventing an identity or a third person.

The same separation applies to policy names. `policy:travel-v1` and `policy:travel-v2` are distinct version entities. The mentions `m-policy-v1`, `m-policy-v1-meeting`, `m-policy-v2-meeting`, and `m-policy-v2-register` are separate textual occurrences referring to those versions. A mention such as "version 2" is meaningful because its source context supplies a policy subject; outside that context the words alone may be ambiguous. Modelling the policy family as though it were identical to either version would blur the temporal question the fixture is designed to answer.

Each source has a logical locator: `R1:P1-P2`, `D1:A1`, or `M1:paragraph-3`. These identify synthetic source regions, not byte offsets or complete source docu-

ments; the construction input retains excerpts and checks character offsets within them; the register locator covers both rows rather than singling out one claim. A production evidence chain should point to a stable source version and a reproducible row key, page and paragraph, or character offsets against a content-addressed text extraction. Offsets without a reproducible text version are fragile; row numbers are fragile if rows can be sorted.

Stable, namespaced identifiers.

Identifiers should be stable and namespaced. The fixture's IDs (`org:northbridge`, `project:atlas`, `policy:travel-v1`, and `person:morgan-lee`) are readable synthetic identifiers; the prefix suggests a kind but does not by itself guarantee global uniqueness or establish who issued it. Real source identifiers should retain their issuing authority, since `1234` in one system may identify something different from `1234` in another. A label change should not silently create a new entity, and a reused label should not silently merge two entities. If two records are linked, preserve the mapping and the basis for it so that a later correction can split the records without rewriting the source history.

The World Wide Web Consortium (W3C) RDF 1.1 Concepts recommendation makes a related distinction between Internationalised Resource Identifiers (IRIs), literals, and blank nodes: identifiers and values play different roles in a graph. Northbridge's JSON schema is not RDF, but the conceptual lesson applies across storage choices. A date such as `2024-03-15` is a value; `project:atlas` is a reference; "Atlas" is a label; and the occurrence `m-atlas` is evidence that a particular source used that label. Retaining these distinctions allows labels and interpretations to change without losing what the source actually contained.

Model qualified claims and source assertions.

A bare triple such as `policy:travel-v1 applies_to project:atlas` omits information needed to answer the Northbridge questions. The fixture's claim `c1` adds source `s-register`, mention `m-policy-v1`, the validity interval from 1 January to 1 April 2024, and positive polarity. These fields make the statement inspectable and temporally queryable. The corresponding positive relation in `relations` is only an analytical projection; it omits source, mention, interval, and polarity and must not become the evidence record used to explain an answer.

Qualified assertions.

The subject–predicate–object form remains useful for the core relation, but many claims are not adequately represented by an unqualified edge. A policy's applicability can be conditional on jurisdiction, staff class, or activity; ownership can have a role and appointment period; an employment relation can be asserted by one source and disputed by another. One representation is to make each qualified assertion a first-class record, as `claims` does, with references to its participants and evidence. Another is an n-ary relation node or a named graph with attached metadata. The design choice should preserve the fields needed by competency questions and validation. W3C's guidance on n-ary relations describes this general modelling problem; it does not prescribe one universal encoding.

Distinguish a proposition from an assertion of that proposition. In this primer, a claim is the qualified proposition under consideration; an assertion is a source or process presenting it. Claim `c4` says that `policy:travel-v2` applies to Atlas from 15 March according to the minutes. Claim `c6` says that the same version applies from 1 April according to the register. Claim `c5` records the minutes' negative assertion that version 1 applies on 15 March. Keeping source-specific assertions means the minutes and register remain separately queryable. It also avoids the misleading operation of overwriting one value with the other.

Temporal scope.

The temporal contract must distinguish three end kinds; nullness alone is not enough. A `bounded` claim has `valid_to > valid_from` and means the half-open interval `[valid_from, valid_to)`. An `unbounded` claim has no `valid_to` and explicitly says that the source asserts continuation; it remains revisable and is not a promise of eternal truth. An `unknown` claim has no `valid_to` but has a `supported_through` date: the source supports the inclusive period `[valid_from, supported_through]`, and applicability afterwards is indeterminate. `supported_through` is the last domain-valid date supported by the assertion, not its publication, ingestion, review, or transaction time. It must not be inferred from a single observation or from a source publication date.

Thus `c1` is bounded, covers 1 January through 31 March, and `c6` begins on 1 April. Existing claims with an unspecified end should be treated as `unknown` unless the source explicitly supplies continuation; do not silently promote them to `unbounded`. The fixture and runnable script require explicit `end_kind`, explicit nulls where applicable, and the corresponding `supported_through` rule. The

predicate itself also needs a definition: `owned_by` means the register's recorded owner in this example, not necessarily the legal owner, budget holder, or operational decision-maker.

Polarity and conflicting evidence.

Polarity is not the same as truth status. `c5` is a negative-polarity assertion from the minutes; its presence records that the minutes deny applicability of version 1 from 15 March. It does not erase positive `c1`, and the fact that `c1`'s validity interval ends on 1 April does not make `c5` true. The positive-claim `relations` projection intentionally excludes `c5`, so analyses over that projection cannot answer questions about the disagreement. Query design must choose whether it needs the positive topology, the full assertion ledger, or both. A projection is a derived view with documented loss, not a replacement for source-backed claims.

5 Build Conservatively and Resolve Identity with Abstention

The claim model specifies what a usable record must retain. Construction must now decide whether a source supplies enough evidence to populate that record and link its participants; missing evidence can require abstention at either step.

5.1 From candidate evidence to an abstaining identity decision

Extraction turns source material into candidate mentions and assertions. It is an interpretation step, even when a rule-based parser is used: sentence boundaries, negation scope, coreference, version references, and date expressions can all be misread. Preserve the input source and version, the extracted text or span, the proposed participants and predicate, and the extraction method. Separate this candidate record from a later accepted assertion so that changed rules can be compared against the same evidence.

The Northbridge examples expose why this matters. The register says version 1 applies to Atlas from 1 January until 1 April; the minutes, dated 15 March, say version 2 already applies and explicitly deny that version 1 applies. A system that extracts only affirmative noun–verb patterns may retain `c4` and lose `c5`. A system that treats the meeting date as the start date of the policy's validity may confuse when a statement was made with when the stated relationship holds.

The fixture encodes the meeting assertions as valid from 15 March, but a richer representation would separately store the minutes' publication or event date and the time referred to by each statement. Extraction must retain enough context to assess those interpretations rather than hide them in a single edge.

Identity resolution.

Entity resolution is a separate decision: given a mention or source record and candidate entities, decide whether the evidence supports identity. Candidate generation and candidate matching have distinct failure modes. A blocking rule may fail to propose a true match; a matcher may incorrectly accept a proposed pair. Statistics Canada's study of blocking errors discusses why missed candidates need separate measurement from match decisions. Names, type compatibility, co-occurring identifiers, dates, and source-specific rules can help rank or reject candidates, but no string similarity threshold supplies evidence that is absent from the source.

The executable identity walkthrough uses the separate retained excerpts and directory rows. The register says "Owner Morgan Lee has directory ID NB-104." Name matching proposes `person:morgan-lee`, whose directory row also has `NB-104`; `resolve` accepts that candidate. The fixture records the corresponding outcome as `r1` for `m-morgan`, although the identifier itself is outside the fixture. A separate test supplies `NB-999` for Morgan and gets a rejection for an incompatible identifier; that test is not an additional fixture decision.

The minutes say "Alex Kim attended." Name matching finds `person:alex-kim-1` (`NB-201`) and `person:alex-kim-2` (`NB-202`), but the sentence supplies neither identifier. The executable abstains with both candidates; curated decision `r2` likewise leaves `m-alex.entity` and its selected candidate null. These deliberately simple exact-match rules demonstrate acceptance, rejection and abstention on synthetic inputs; they do not measure identity-resolution accuracy or prove that an actual person attended.

The fixture records each decision ID, mention, candidate set, outcome, selected candidate where applicable, and reason. It contains one acceptance and one abstention, but no rejection. An operational decision record should also identify the method and version, review status and decision time. The Python validator checks the decision references and selected-candidate rules, and asserts the Alex abstention; it does not run a matcher or write the decisions into DuckDB.

Abstention and review.

Abstention is useful because the costs of errors are asymmetric. A missed link may leave a query incomplete; a false identity merge can attribute meeting attendance or policy ownership to the wrong person and contaminate every downstream path. Preserve uncertain candidate links outside the accepted graph, and make review possible. Pairwise decisions also need global consistency checks: accepting A–B and B–C does not prove A–C if the evidence is inconsistent or entities are not governed by a strict equivalence relation. Record enough provenance to reverse or revisit a merge.

Do not interpret a model's score as a probability of real-world truth unless it has been calibrated and evaluated for that task and population. Keep uncertainty tied to its source: ambiguous name, uncertain date, source reliability, incomplete coverage, or extraction disagreement. Evaluation should report candidate-generation recall separately from accepted-match precision and review/abstention rates, using an explicitly described labelled sample. The synthetic fixture is a behavioural example; it cannot establish those rates for Northbridge or any wider population.

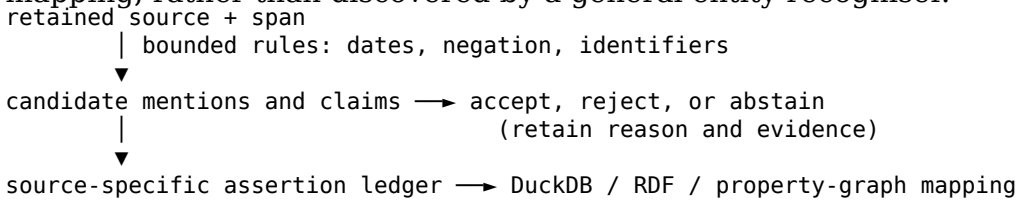
5.2 Bounded Source Claim Construction

Now that mentions, identities and abstention rules are explicit, construction can populate the qualified claim model without silently turning text into truth. The companion construction implementation reads versioned synthetic excerpts. Its three sentence patterns produce candidates with subject, predicate, object, source, dates, end kind and polarity, plus an evidence locator, character offsets, exact matched text and method `northbridge-patterns-v1`. Offsets are checked against the retained text. The following table follows the actual source sentences into candidates; the dates are calendar dates.

Retained text and locator	Candidate and interpretation	Ledger mapping
`R1:P1-P2`: "Travel Policy v1 applies to Atlas from 2024-01-01 until 2024-04-01."	Version 1 `applies_to` Atlas, positive; `bounded`, starting 1 January and ending exclusively on 1 April.	Constructed `c1`, with mention `m-policy-v1`.

Retained text and locator	Candidate and interpretation	Ledger mapping
`M1:paragraph-3`: “On 2024-03-15, version 2 applies to Atlas.”	Version 2 `applies_to` Atlas, positive; `unknown` end, supported through 15 March.	Constructed `c4`, with `m-policy-v2-meeting`.
`M1:paragraph-3`: “On 2024-03-15, version 1 does not apply to Atlas.”	Version 1 `applies_to` Atlas, negative; `unknown` end, supported through 15 March.	Constructed `c5`, with `m-policy-v1-meeting`.
`S1:1`: “Travel Policy v1 applies to Atlas from 2024-01-01 and continues until further notice.”	A separate synthetic positive candidate with `unbounded` end because the source explicitly asserts continuation.	Used in a separate classifier test; not inserted as a ledger claim.

For the meeting sentences, the rule encodes the stated date as `valid_from` and `supported_through`; it does not infer continuation beyond that date. The method is deliberately restricted to these sentence forms and policy versions. Its selected claim IDs and mention IDs are assigned by a fixed source-and-subject mapping, rather than discovered by a general entity recogniser.



The executable builds source-derived `c1`, `c4`, and `c5` from the retained excerpts, keeps their evidence spans in an `extraction_evidence` table, and merges them with the curated remainder of the ledger before loading DuckDB. It also compares the current constructed rows with the fixture as a diagnostic, rather than requiring the fixture to be edited before a changed source can be built. The correction demonstration later in the primer goes further: it rebuilds a fresh database from a changed source excerpt, reruns the affected question, and asserts the changed answer. The other ledger claims are curated: `c2` records ownership, `c3` and `c7` project affiliation, and `c6` the register’s version 2 applicability. They are not extracted by this slice, and their complete source passages are

not retained. The constructed-versus-curated distinction describes how these example rows were supplied, not a judgement that either is independently verified. Full-document optical character recognition, open-domain language understanding, learned matching, and a general update engine remain outside this milestone.

5.3 Work one claim from source to answer

The whole construction can be understood without a parser or graph service. Take the sentence “On 2024-03-15, version 1 does not apply to Atlas.” The sentence is not yet a graph edge. It is a source occurrence with a location, wording, and an interpretation that still needs to be recorded.

Stage	Stored or decided result	Why it matters
Source occurrence	`m-policy-v1-meeting`, in the Atlas minutes at `M1:paragraph-3`	A later reviewer can find the exact sentence and distinguish it from another mention of version 1.
Mention interpretation	“version 1” refers to `policy:travel-v1`; “Atlas” refers to `project:atlas` in this meeting context	The model records an identity decision instead of pretending that labels are globally unique.
Qualified claim	`c5`: `policy:travel-v1` `applies_to` `project:atlas`, negative polarity, valid from 15 March, end kind `unknown`, supported through 15 March	The relationship can be queried without discarding its date or the fact that the source denies it.
Source assertion	The minutes present `c5`; the register separately presents positive `c1` for 1 January–1 April	Competing evidence remains visible rather than being overwritten by a preferred value.

Stage	Stored or decided result	Why it matters
Reader question	“What does the evidence say about version 1 on 15 March?”	The question fixes the date and prevents a timeless answer from being smuggled in.
Result	`c1` is applicable and positive; `c5` is applicable and negative; answer policy returns `abstain: conflicting evidence`	The result states what the selected evidence supports and why it does not choose a winner.

Now add the other path edge: the directory says Atlas belongs to Northbridge. The reader can ask whether there is a path from version 1 to Northbridge using the sequence `applies_to → belongs_to`. The answer is a two-hop path whose first hop is `c1` and whose second hop is `c3`. It is not a new claim that version 1 applies to Northbridge. The path is a useful answer to a structural question, but its interpretation remains bounded by the meaning of each predicate and by the dates and sources on each hop.

This distinction becomes important when the question changes. “Is there a path?” and “Was the policy applicable to the organisation on 15 March?” are not equivalent. The first may be answered by following the selected positive topology. The second must classify each hop at the requested date, retain negative and indeterminate evidence, and state whether the policy allows that path to support an organisation-level conclusion. A graph can make the first answer easy while leaving the second unresolved; that is a modelling result, not a failure of graph traversal.

Finally, consider the minutes’ mention of “Alex Kim”. The source occurrence survives, but two directory entities have that label and the sentence provides no distinguishing identifier. The graph therefore stores an unresolved mention, both candidates, and an abstention. It does not create an edge from the meeting to either person. A query can report that the meeting contains an unresolved Alex Kim mention; it cannot responsibly answer which directory person attended. The absence of an edge here is not a negative attendance claim. It is the result of an explicit identity decision.

These three outcomes—an evidence-backed direct assertion, a qualified path, and an unresolved mention—are the basic reader operations for the rest of the primer. In each case the graph is useful because the intermediate decision is available for inspection. In each case the graph is limited because the answer depends on source scope, predicate meaning, temporal policy, and completeness.

6 Preserve Provenance, Time, Conflict, and Uncertainty

These concerns determine whether a graph answer can be traced, dated, compared with competing evidence, and qualified when the evidence is incomplete. Treat them as connected controls: provenance identifies the assertion, time limits its scope, conflict preserves disagreement, and uncertainty constrains the answer.

6.1 Keep the evidence chain interpretable

Provenance makes an answer auditable: which source version supported which assertion, what activity extracted or reviewed it, and which release and selection rule produced the answer. The W3C PROV Ontology (PROV-O) vocabulary models entities, activities, and agents and can inform an implementation. Even without adopting PROV-O, preserve stable source identifiers, version or digest, source and ingestion dates where known, reproducible locator, extraction or review method, and the responsible process or agent. A source publisher, the person quoted, and the extractor are different roles and should not be collapsed.

For the Northbridge register, one useful provenance chain is `s-register` as the source entity, an extraction activity that records the parser version and retained span, a review activity that accepts or changes the candidate claim, and a named process or agent responsible for that activity. A later release can then say that `c1` was derived from the register's `R1:P1-P2` span by one extraction run and included after one review rule, without implying that the extractor authored the policy. If the register is corrected, retain a new source version and connect the changed claim to the new activity; do not rewrite the old claim and lose the explanation of what an earlier release contained.

This distinction answers different audit questions. "Which document did the

answer come from?" needs the source and locator. "How did this claim enter the graph?" needs the extraction or review activity and method version. "Who may approve its publication?" needs an accountable agent and release record. A checksum proves which bytes were processed, but it does not prove that the source was authoritative or that the extraction rule interpreted it correctly.

Valid time and observation time.

Time commonly has at least two axes. Valid time describes when the asserted relationship holds in the domain; transaction or observation time describes when the system recorded that assertion. The fixture's `valid_from` and `valid_to` encode valid time, while its `sources[].published` gives source publication dates. It has no ingestion timestamp or transaction history, so it cannot answer "what had the system recorded by 20 March?" without an external release record. A corrected register imported on 1 May should not rewrite the evidence available in an earlier release. Preserve both the old and new source versions and state whether a historical query asks what was known then or what current evidence says about then.

Consider `c1`, which is valid from 1 January until 1 April. A query asking whether version 1 applied on 31 March can use its valid interval; a query asking what the graph knew on 20 March also needs a release or ingestion timestamp showing that `c1` had already been admitted. A source published on 10 March might be observed by the system on 12 March, reviewed on 14 March, and published in graph release 3 on 16 March. Those dates answer different questions. If a corrected source arrives on 1 May, a current query about 31 March may use the correction, while a reconstruction of release 3 must use only the evidence admitted by that release.

When a schema stores only `valid_from` and `valid_to`, it can answer a domain-time question but cannot reconstruct the graph's historical knowledge. That is not a reason to invent an observation date from the source publication date. Add the missing release history or state that the historical query is unsupported.

Conflict and authority.

For the date 15 March, the register's `c1` covers Atlas under version 1; the minutes' `c4` asserts version 2 and `c5` denies version 1 for the same project and date. Those are incompatible assertions under the fixture's intended interpretation. `c6` begins version 2 on 1 April and is consistent with the end of `c1`; it is

not itself a contradiction. Before labelling two records contradictory, compare subject, predicate, object or competing value, temporal scope, jurisdiction, and relevant conditions. A conflict requires overlapping scope and mutually exclusive claims, not merely two different records.

Authority is purpose-specific. To answer “what did the register record?”, use the register. To answer “what did the minutes say?”, report the minutes. To answer “which version should a user treat as governing?”, the fixture alone cannot settle legal or operational force; its teaching rule can prefer the register only for the narrower question about recorded policy applicability. Keep the losing assertion visible and state the preference rule. Otherwise a source ranking can look like a truth mechanism and conceal a material disagreement.

Uncertainty and path interpretation.

Uncertainty has multiple causes and should not be compressed into one confidence number. The Alex identity is underdetermined because the source lacks a distinguishing identifier. The applicability disagreement is not an extraction uncertainty in the fixture: two explicit, incompatible records are present. Source completeness is unknown because only three synthetic source records are included. An unknown end on `c2` means duration is supported only through its retained horizon; it is not an eternal interval. Each condition changes what a query can conclude and should be represented or reported separately.

Agreement between two assertions is not automatically independent corroboration. They may copy the same upstream register, share an author, or be generated from one database. Provenance can reveal common origin; counting edges cannot. Similarly, a path from a policy through a project to an organisation proves only that the selected projection contains those edges. It does not establish causality, authority, or completeness. Return evidence for each hop and keep interpretation distinct from topology.

7 Privacy and Proportionate Identity

Entity resolution can turn dispersed records into a more identifying profile than any single source. A name, project affiliation, and meeting attendance may be low-risk in isolation but sensitive when linked and made searchable. The Northbridge fixture is fictional, yet its two Alex Kim records demonstrate a practical harm: an unsupported merge could attribute attendance to the wrong person.

In a real system, that same error can affect access decisions, employment records, or investigations.

Collect and expose only the identifiers and attributes required by the competency questions. If a policy query needs the owner relation, it may not need personal contact details or a broad employee profile. Keep direct identifiers in a controlled mapping where possible, apply access controls to source text and derived claims, and record retention and deletion rules. A graph edge can still identify a person when linked to other records; changing its storage format does not remove that concern. The UK government's data protection guidance describes the principle that personal data use should be relevant and limited to what is necessary. This fictional design does not select a legal basis or retention period; those decisions require review for the actual jurisdiction and purpose.

For Q5, assume two audiences and one purpose: an ordinary policy-query user needs to know whether attendance can be attributed, while an authorised source reviewer may inspect the ambiguity. Return to the ordinary user only "the minutes mention an attendee named Alex Kim; identity unresolved", the meeting date and a non-sensitive source class. Do not expose either directory ID, candidate identifier, full minutes text, or a person-to-meeting edge. The reviewer may see `m-alex`, `r2`, both candidate IDs and their controlled directory identifiers, the exact `M1:paragraph-3` span, and the reason for abstention, subject to access to those source records. If that access is absent, report that evidence is restricted rather than reveal it indirectly through a traversal. This is an illustrative access decision, not an enforced permission rule in the executable. Both audiences receive an abstention on identity until an authorised distinguishing source is reviewed.

Privacy also affects provenance design. A public answer may need to show a source class, version, and redacted locator rather than expose an entire restricted document. The system should preserve enough controlled evidence for authorised audit while preventing broader access through graph traversal. Test both what an authorised reviewer can trace and what an ordinary query user can infer. Access restrictions and redaction must not silently erase the fact that evidence was withheld; report the limitation at a level that does not itself disclose protected information.

The practical design test is whether each identity link and exposed attribute is necessary for a stated question, supported by evidence, and available to the

intended audience. For the ambiguous Northbridge attendee, abstaining protects both correctness and the people represented. It also leaves a clear route for adding a distinguishing identifier later, should an authorised source provide one.

Apply the same decision pattern to every answer that crosses an audience boundary:

1. State the question and the minimum fact needed to answer it. For an ordinary user asking whether attendance can be attributed, the minimum is that the minutes mention an unresolved Alex Kim; neither candidate's directory identifier is needed.
2. Separate evidence needed to justify the answer from attributes that would merely make a person easier to identify. An authorised reviewer may need `m-alex`, `r2`, both candidates and the exact source span to investigate the abstention, while a general user does not.
3. Check whether the evidence is available to that audience. If the minutes are restricted, report that the supporting evidence is unavailable to the requester; do not disclose a candidate indirectly through a graph path or a count that reveals the hidden choice.
4. Return the least identifying answer that remains useful, and record the access decision and policy revision for later review. A later authorised source can resolve the mention without rewriting the original occurrence or exposing the earlier restricted evidence to everyone.

This pattern also exposes failure cases. Returning both candidate IDs to a general user fails minimisation even if the identity decision correctly abstains. Returning only "unknown" to an authorised reviewer fails auditability because it hides the evidence and decision that could be challenged. Treating an access denial as "no matching person" fails correctness because it converts restricted evidence into a false negative. Privacy therefore changes the answer contract and the evidence path, not just the columns selected from a database.

8 Use the Graph: From Claims to Interpreted Queries

The preceding decisions become useful when a reader can ask a question and follow the answer back through the claims that support it. Start with the reader-facing operation, not with a database product. A direct question returns the

relevant assertions; a path question returns every hop and its evidence; a dated question classifies each assertion before combining the results; and a conflict or uncertainty policy determines whether the final answer should be supported, refuted, or withheld. The same reasoning can be implemented with tables, RDF, a property graph, or a service API.

The small example below uses a tabular ledger as an implementation companion. It is intentionally not the source of the explanation: the tables and result interpretations above and below are the standalone teaching route. The code and schema show one reproducible implementation of the decisions, while the reader should not need to inspect repository files to understand what the queries mean.

The canonical runnable companion constructs source-derived claims, merges them with explicitly curated rows, reports diagnostic drift from the teaching fixture, validates references and intervals, writes claims and provenance to a file-backed DuckDB database, closes it, reopens it read-only, and runs bounded queries with assertions. It is not a general source extractor. The repository keeps the exact container command and example manifest separately so that this standalone explanation does not make its reader learn a file layout before understanding the model.

The runner provides a fresh temporary output directory, disables network access and discards the database afterwards. The script uses `EXAMPLE_OUT` when set and otherwise writes to its working directory. It creates tables with plain `CREATE TABLE`, so running it again against the same output database fails rather than silently replacing the previous build. Use a fresh directory for each standalone run. DuckDB documents file-backed and read-only connections; persistence here means the script can reopen and query its file, not that it publishes an enduring release.

To test a changed retained source without editing the fixture, provide a copy of the source input through `SOURCE_RECORDS_PATH` and a fresh `EXAMPLE_OUT` directory. The script builds the source-derived rows, loads and queries them, reports which fixture rows drifted, and skips only the frozen current-fixture answer assertions when drift is deliberate. Structural validation and the source-evidence joins still run. For example, changing the register's end date to 1 March produces a rebuilt `c1` with that exclusive end and returns `refuted` for the 15 March question because the applicable minutes assertion is negative. This is a bounded

source-edit demonstration, not a general ingestion or migration tool.

The fixture contains eight entities, three source records, seven mentions and seven claim records. It also contains five `relations` for a separate analytical projection. The Python script validates entity references but does **not** create an entities or relations table. It writes `claims`, `sources`, `mentions` and `extraction_evidence`; the last table stores locator, character span, exact text and method for source-derived `c1`, `c4` and `c5`, linked by claim ID. Curated claims have no extraction-evidence row. This supports evidence-bearing joins for the constructed slice, but an application needing entity labels, types, and database-enforced references must add those tables and load them. The R example reads entities and the precomputed positive `relations` projection from `fixture.json`; it does not consume the rebuilt DuckDB database or a changed `SOURCE_RECORDS_PATH` input. The retained source records are a separate synthetic input and include a deliberately refused date phrase. If those records change, the fixture's curated fields and positive projection need an explicit rebuild or consistency check before the R output can be treated as synchronised.

The executable assertions also give a compact account of dated answers and evidence. The first four rows ask whether version 1 applies to Atlas across both register and minutes; the last two classify both edges of the register-backed version 1 → Atlas → Northbridge path. Locators are joined from the source table, and mention text is available for `c1` and `c5`.

Date and question	Evidence and temporal classification	Executable result
1 March 2024, applicability	`c1` positive, applicable (`R1:P1-P2`); `c5` outside (`M1:paragraph-3`).	`supported`
15 March 2024, applicability	`c1` positive and `c5` negative, both applicable, with their respective register and minutes locators.	`abstain: conflicting evidence`
16 March 2024, applicability	`c1` applicable; `c5` indeterminate beyond its support horizon.	`abstain: indeterminate temporal scope`

Date and question	Evidence and temporal classification	Executable result
1 April 2024, applicability	`c1` outside at its exclusive end; `c5` indeterminate.	`abstain: indeterminate temporal scope`
1 January 2024, two-hop path	`c1` applicable (`R1:P1-P2`); `c3` applicable (`D1:A1`).	`applicable`
1 March 2024, two-hop path	`c1` applicable; `c3` indeterminate beyond 1 January.	`indeterminate`

These are fixture regression results under the executable's conservative answer policy, not a legal determination or evidence of production accuracy. The path status is a classification of both premises, not an inferred organisational claim.

8.1 Companion implementation: a tabular evidence ledger

In the target ledger, each JSON claim is a source-specific assertion with an ID, subject, predicate, object, source, optional mention, valid interval, temporal end kind, optional support horizon, and polarity. The field contract is: `bounded` requires `valid_to` and no `supported_through`; `unbounded` requires both to be null and explicit source wording of continuation; `unknown` requires a null `valid_to` and `supported_through >= valid_from`. Dates use calendar-date strings. The horizon is the last supported domain-valid date, inclusive; it is not publication or ingestion time. The checked-in fixture uses this contract. In either form, `c1` says that `policy:travel-v1` applies to `project:atlas` from 1 January to 1 April 2024, according to `s-register`; `c5` records a negative assertion about the same subject, predicate and object from `s-minutes`, starting on 15 March. The two rows preserve disagreement. `c4` and `c6` both assert that version 2 applies to Atlas, but they have different sources and start dates. Neither should be collapsed into a single source-free edge if the question concerns evidence.

The companion ledger has four conceptual tables. `claims` holds the subject, predicate, object, source, optional mention, valid interval, end kind, support horizon, and polarity. `sources` holds source titles, publication dates, and logical locators. `mentions` preserves the source occurrence and any resolved entity. `extraction_evidence` links a constructed claim to its locator, character span,

exact text, and extraction method. The schema is deliberately small; the important lesson is the fields and their relationships, not the syntax of a particular data-definition language.

The executable loads both qualifier fields and validates the three combinations before insertion. A production-aligned schema would also make `end_kind` non-null, add keys and foreign keys, and enforce the cross-field combinations in the database as well as in application code.

Loading uses parameterised `executemany` calls, with columns selected in the same order as the DDL. The JSON dates are converted to DuckDB `DATE` values on insertion. Prepared values keep data separate from SQL syntax; they do not turn a dynamic table or column name into a safe identifier. See the DuckDB Python DB-API documentation for parameter binding. The executable validates before insertion, rather than relying on the DDL for uniqueness, required values or foreign keys.

The example's DDL does not enforce keys, required fields or foreign keys. The Python validator checks several of these before insertion, including that a claim's mention comes from its source and identifies a participant. A more defensive database schema could add primary and foreign keys, non-null fields and interval checks; cross-row conditions and domain rules would still need additional validation. Consult DuckDB constraints for the pinned version's behaviour.

The source table includes the fixture's logical locator, but the fixture has no source version, digest, ingestion time or complete source document. The separate construction input has a synthetic version and excerpts, which are not loaded into that table. `extraction_evidence` records spans of those excerpts for three claims, not immutable source bytes or an independently verified document. A production system should add source versions and use assertion-to-evidence links where one assertion depends on several passages. Evidence links can share a source and are not necessarily independent corroboration.

The distinction between a proposition and an assertion matters when several records say the same thing. This fixture uses `claims` as source-specific rows: the source and interval are part of each row. A larger model may instead give a canonical proposition its own ID and record each source assertion separately, including its extraction method and review state. Either design can work if it

preserves which source made which assertion, under which scope, and does not mistake a merged proposition for a verified fact. The W3C PROV data model provides a general vocabulary for entities, activities and agents when interoperable provenance is needed; the fixture does not implement PROV.

8.2 Direct questions, paths, and evidence

A direct query returns all recorded assertions for a subject and predicate. For `policy:travel-v1` and `applies_to`, the result contains `c1` with positive polarity and `c5` with negative polarity. The operation deliberately has no date or source filter. It establishes that the ledger contains two differently signed assertions; it does not decide which applies on a particular day.

Filtering to positive rows first would hide the negative evidence. Filtering to the register answers a different, explicitly source-scoped question, not an unrestricted question about all evidence.

A two-hop query follows a known relationship sequence. It joins the object of a positive `applies_to` claim to the subject of a positive `belongs_to` claim.

With parameters `policy:travel-v1`, `applies_to` and `belongs_to`, the result is `(c1, c3, org:northbridge)`: the policy is linked through Atlas to Northbridge. Both claim IDs are returned, so the evidence for each step can be inspected. The join establishes a path under the query's conditions; it does not create a new assertion that the policy applies to the whole organisation. The executable's two-hop query has no time filter and therefore does not establish that both links hold on any requested date. For a dated path, apply the same valid-time predicate to **each** row, and retain the individual intervals and sources. If several assertions support either step, the join can yield several paths; deciding whether to deduplicate answers is separate from retaining their evidence.

Bounded predicates and hop counts also constrain cost and interpretation. An unrestricted traversal may find a connection through irrelevant relationship types, combine incompatible dates, or return many paths created by duplicated evidence. State which predicate sequences answer the question and why their direction matters. A path through `owned_by` has a different meaning from one through `belongs_to`, even if both lead to a familiar name.

8.3 Valid Time, Conflict and Evidence

The target query policy classifies each *validated* matching assertion before aggregating an answer. A bounded end is exclusive; an explicit unbounded end remains applicable after its start; an unknown end is applicable only through its inclusive support horizon and is indeterminate afterwards. In prose, the classification is: before `valid_from`, outside; inside a bounded interval, applicable; after a bounded interval, outside; after an explicit unbounded start, applicable; inside an unknown assertion's support horizon, applicable; and after that horizon, indeterminate. Any unrecognised end kind is invalid.

The source-specific answer policy is deliberately conservative: conflicting applicable polarities produce `abstain: conflicting evidence`; any indeterminate row produces `abstain: indeterminate temporal scope`; one applicable polarity produces `supported` or `refuted`; and no applicable or indeterminate evidence produces `abstain: no evidence`. This is a teaching policy, not the only valid aggregation rule. Do not turn an unknown negative into a definite refutation, or discard it to obtain a definite support. The executable applies this policy after retaining each row's temporal classification. The `ELSE 'invalid'` branch catches an unrecognised end kind that reaches this expression; it does not validate arbitrary malformed rows. Python rejects the fixture's tested date-field combinations before insertion, and the query raises if a returned row is classified `invalid`. Missing or null fields outside that validated input contract need separate schema checks; a SQL `NULL` comparison can bypass the apparent branch outcome.

Under this contract, 31 December 2023 has no evidence; 15 March has applicable `c1` and `c5` and therefore a conflict; 16 March has applicable `c1` and indeterminate `c5`; and 1 April and 15 April have an outside `c1` and indeterminate `c5`. These are classifications relative to the selected evidence and policy, not claims that the world changed at an unobserved time. They do not establish when a reader could first have known each statement. For that, record when each assertion became available to the system, preserve its changes and filter both valid time and knowledge time. Publication date alone is not a substitute for ingestion or transaction history.

Here `refuted` means that the selected applicable rows contain negative evidence and no positive evidence under this policy. It is not a judgment that the negative source is authoritative or that every other relevant record has been found. No

evidence is different from explicit negative evidence. Nor does a conflict force a particular authority decision: the policy does not rank the register above the minutes.

A provenance query joins each matching claim to its source title and locator, then left-joins its mention text when one exists. It returns both `c1` and `c5` with their different source labels and locators, rather than returning only the selected answer.

The executable returns `c1` with `Authorised policy register`, `R1:P1-P2` and `Travel Policy v1`, and `c5` with `Atlas meeting minutes`, `M1:paragraph-3` and `version 1`. The left join matters because a claim such as `c7` has no mention. These source-level locators are logical labels, not the underlying documents or verified quotations. A real system needs source versions, reproducible locations, access conditions and retrieval context.

A dated query for version 2 illustrates a separate issue: `c4` is positive from 15 March and `c6` is positive from 1 April. The presence of two positive rows after 1 April does not prove independent corroboration or settle the register-versus-minutes interpretation. If an application adopts an authority hierarchy, make that policy a versioned input to answer generation, retain excluded evidence, and explain why it applies to this predicate, scope and date. A later correction must be able to change the answer without erasing the historical disagreement.

9 Optional Representations: RDF, Property Graphs, and Projections

A representation is useful when it preserves the distinctions needed to answer a question, not merely when it draws a convincing network. Follow the Northbridge register's version 1 applicability statement through each representation: the policy version applies to Atlas from 1 January 2024 up to, but excluding, 1 April; the assertion comes from register `R1`, locator `R1:P1-P2`. The meeting minutes make a competing statement about version 2 on 15 March. A representation that retains only two policy-to-project arrows cannot reconstruct the dates, the source of either statement, or why the statements conflict. Nor should it turn the unresolved meeting mention of Alex Kim into an identified person.

This reference-only branch starts with the same evidence requirements as

the DuckDB route and asks how another representation could preserve them. RDF readers can follow triples through querying, validation, and provenance; property-graph readers can proceed to Property Graphs and Cypher. The comparison and storage subsections apply to either choice, while the Neo4j smoke test is a prospective product-specific branch. These examples are not a second executable construction route; choose that scope deliberately before committing to an RDF or property-graph serving layer. Resume the main route at Analyse a Projection in R.

The examples in this section are illustrative designs inspired by the fixture. Their identifiers and vocabulary are local to the examples; they are not a claim that a particular RDF export, property-graph adapter, or Neo4j service has been implemented. In particular, distinguish a *claim*, which describes a qualified proposition, from an *assertion*, which records a source or process presenting that proposition. A source can support several claims, and several assertions can refer to the same claim while retaining separate locators and histories.

9.1 RDF: Triples and Qualified Claims

The Resource Description Framework (RDF) represents a graph as a set of triples: subject, predicate, and object. A triple says that the predicate relates the subject to the object. The subject is an IRI or blank node, the predicate an IRI, and the object an IRI, blank node, or literal. An IRI is an identifier in a global naming system; a literal is a value with a datatype or language tag. Blank nodes identify resources without providing a globally meaningful IRI. A graph is a *set*, so repeating the same triple does not record a second independent source. See RDF 1.1 Concepts for the abstract model.

For example, the Turtle statement `ex:atlas ex:belongsTo ex:northbridge .` is a readable serialisation of one triple. Prefix `ex:` abbreviates a namespace, rather than defining a separate kind of identifier. Turtle's punctuation lets multiple triples share a subject or predicate. Two serialisations with different ordering can describe the same RDF graph, and the spelling of a blank-node label in one file is not a stable identifier to use across releases. A persistent source locator or claim that must be cited across imports is therefore better given a governed IRI than identified by a local blank-node label.

A single direct triple such as `ex:travelV1 ex:appliesTo ex:atlas` does not contain the interval or source. Putting such a triple in a graph also asserts

that relation in that graph; the existence of an assertion *about* the relation is a different modelling choice. For the conservative temporal contract, map `valid_from` and `valid_to` to `ex:validFrom` and optional `ex:validTo`; put exactly one `ex:endKind` on the claim using `ex:BoundedEnd`, `ex:UnboundedEnd`, or `ex:UnknownEnd`; put `ex:supportedThrough` on the source-specific assertion; and put required `ex:polarity` on the qualified claim. Use `xsd:dateTime` at midnight Coordinated Universal Time (UTC) for these domain dates so portable comparisons in the SPARQL Protocol and RDF Query Language (SPARQL) use one ordered datatype; this is an encoding convention, not invented time-of-day precision. Model the qualified proposition as its own resource instead:

```
@prefix ex:   <https://example.org/northbridge/> .
@prefix xsd:  <http://www.w3.org/2001/XMLSchema#> .
@prefix prov: <http://www.w3.org/ns/prov#> .

ex:claim-v1-atlas a ex:ApplicabilityClaim ;
    ex:policyVersion ex:travelV1 ;
    ex:project ex:atlas ;
    ex:validFrom "2024-01-01T00:00:00Z"^^xsd:dateTime ;
    ex:validTo "2024-04-01T00:00:00Z"^^xsd:dateTime ;
    ex:endKind ex:BoundedEnd ;
    ex:polarity true .

ex:assertion-r1-p1 a ex:Assertion, prov:Entity ;
    ex:asserts ex:claim-v1-atlas ;
    ex:sourceRecord ex:r1 ;
    ex:locator "R1:P1-P2" .

ex:claim-v1-atlas-minutes a ex:ApplicabilityClaim ;
    ex:policyVersion ex:travelV1 ;
    ex:project ex:atlas ;
    ex:validFrom "2024-03-15T00:00:00Z"^^xsd:dateTime ;
    ex:endKind ex:UnknownEnd ;
    ex:polarity false .

ex:assertion-minutes-p3 a ex:Assertion ;
    ex:asserts ex:claim-v1-atlas-minutes ;
    ex:sourceRecord ex:minutes ;
    ex:locator "M1:paragraph-3" ;
    ex:supportedThrough "2024-03-15T00:00:00Z"^^xsd:dateTime .
```

The local vocabulary specifies that `validTo` is exclusive; RDF does not infer that convention from its name. The code expresses a claim resource and an assertion resource; it deliberately does not add the unqualified direct `appliesTo` triple. The assertion's source, locator and support horizon make the example traceable at the fixture's source-level logical granularity. The unknown-ended minutes claim is not applicable after its horizon; it is indeterminate. A production record also needs the retained source version, a reproducible locator, and any extraction or review information required for an audit. Here `ex:r1` is an identifier

for the register record, not the literal bytes of the source. These snippets are illustrative and unexecuted.

This is an n-ary relation: the application of a version to a project depends on more than the two endpoints. The claim resource makes the period and other conditions participants in one identified description. If a second source independently reports the *same* version, project, period and conditions, it can have a second assertion pointing to that claim. If it states a different interval or version, create a distinct claim. The W3C guidance on n-ary relations presents this family of patterns. Claim identity and deduplication remain application decisions; sharing an endpoint pair is not enough to establish that two claims are identical.

RDF datasets can organise a default graph and named graphs. A named graph can group an export or source-specific statements, but its name alone does not automatically say who asserted each statement, when it was obtained, or whether it is authoritative. Publish those semantics and provenance explicitly. Grouping by source may help access or retrieval, but one source can contain several locators and claims, and one assertion may rely on multiple source fragments. Named graphs do not remove the need to model that detail when a question requires it.

The examples use RDF 1.1 constructs for portability. Other statement-annotation facilities exist, but a deployment must specify the exact RDF version, syntax, semantics, and store support before depending on them. Even then, annotation syntax does not decide whether two observations are independent, which source takes precedence, or whether a date has been interpreted correctly.

9.2 SPARQL Evidence Queries

The SPARQL Protocol and RDF Query Language (SPARQL) matches graph patterns and returns bindings for variables. Each triple pattern below must match; the shared variables join the claim to its assertion and source. For the illustrative data above, this query retrieves the register claim applicable on 1 March, including its locator:

```
PREFIX ex:  <https://example.org/northbridge/>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT ?claim ?version ?assertion ?source ?locator ?start ?end
       ?kind ?polarity ?supportedThrough ?temporalStatus
```

```
WHERE {
  ?claim a ex:ApplicabilityClaim ;
         ex:policyVersion ?version ;
         ex:project ex:atlas ;
         ex:validFrom ?start .
  ?assertion ex:asserts ?claim ;
             ex:sourceRecord ?source ;
             ex:locator ?locator .
  ?claim ex:endKind ?kind ; ex:polarity ?polarity .
  OPTIONAL { ?claim ex:validTo ?end }
  OPTIONAL { ?assertion ex:supportedThrough ?supportedThrough }
  BIND ("2024-03-01T00:00:00Z"^^xsd:dateTime AS ?day)
  BIND (IF(?day < ?start, "outside",
       IF(?kind = ex:BoundedEnd,
          IF(BOUND(?end) && ?day < ?end, "applicable", "outside"),
       IF(?kind = ex:UnboundedEnd, "applicable",
       IF(?kind = ex:UnknownEnd && BOUND(?supportedThrough) &&
             ?day <= ?supportedThrough, "applicable",
       IF(?kind = ex:UnknownEnd, "indeterminate", "invalid")))))
       AS ?temporalStatus)
  FILTER (?source = ex:r1)
}
ORDER BY ?version ?assertion
```

The `BIND` classifies evidence under the stated end-kind contract rather than treating an unbound end as open. Its `"invalid"` value is not a general malformed-record detector: a missing required triple can prevent the claim from matching at all, and an absent optional value can be classified by an earlier branch. Validate the shape and date combinations separately. A query that returns only `?temporalStatus = "applicable"` is a deliberately narrower register-selection query; an answer-producing query must retain `indeterminate` rows for conservative aggregation. The source filter asks the narrower question of what the register records; removing it could expose both register and minutes claims, provided both have been loaded with comparable dates. To answer what the meeting said, select the minutes and their locator rather than changing the source record or deleting its competing assertion. The query is an illustrative, unexecuted SPARQL design; check syntax and store support before relying on it.

`OPTIONAL` preserves a solution when the optional pattern has no match, leaving its variable unbound. If multiple ends or multiple matching assertions are present, the query can return multiple rows. That is useful evidence of multiplicity but can also duplicate apparent answers; impose appropriate constraints or aggregate deliberately, without suppressing distinct evidence. An omitted assertion fails the mandatory pattern and disappears from this particular result, so a separate validation or audit query is needed to find unsupported claims.

`FILTER NOT EXISTS` can ask whether a matching pattern is absent from the active dataset. It cannot prove that a source or relationship does not exist in the world. A query's dataset, default graph configuration, entailment settings, and access permissions all affect its results. State whether inferred triples are included and whether the selected release contains all relevant sources before treating absence as meaningful. SPARQL 1.1 defines graph patterns, optional matches, filters, datasets, and result forms; an implementation may add features whose behaviour should be documented separately.

9.3 Validate Graph Structure with Shapes Constraint Language (SHACL)

The Shapes Constraint Language (SHACL) expresses conditions over an RDF *data graph* in a separate *shapes graph*. A validator reports whether the chosen data conforms and identifies violations. For the local modelling rule that each applicability claim has one version, one project, and one start date, a compact shape is:

```
@prefix ex:  <https://example.org/northbridge/> .
@prefix sh:  <http://www.w3.org/ns/shacl#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

ex:ApplicabilityShape a sh:NodeShape ;
    sh:targetClass ex:ApplicabilityClaim ;
    sh:property [ sh:path ex:policyVersion ;
                  sh:minCount 1 ; sh:maxCount 1 ; sh:nodeKind sh:IRI ] ;
    sh:property [ sh:path ex:project ;
                  sh:minCount 1 ; sh:maxCount 1 ; sh:nodeKind sh:IRI ] ;
    sh:property [ sh:path ex:validFrom ;
                  sh:minCount 1 ; sh:maxCount 1 ; sh:datatype xsd:dateTime ] ;
    sh:property [ sh:path ex:validTo ;
                  sh:maxCount 1 ; sh:datatype xsd:dateTime ] ;
    sh:property [ sh:path ex:endKind ;
            sh:minCount 1 ; sh:maxCount 1 ; sh:in ( ex:BoundedEnd ex:UnboundedEnd ex:UnknownEnd ) ] ;
    sh:property [ sh:path ex:polarity ;
                  sh:minCount 1 ; sh:maxCount 1 ; sh:datatype xsd:boolean ] ;
    sh:property [ sh:path [ sh:inversePath ex:asserts ] ;
                  sh:minCount 1 ; sh:class ex:Assertion ] .

ex:AssertionShape a sh:NodeShape ;
    sh:targetClass ex:Assertion ;
    sh:property [ sh:path ex:sourceRecord ; sh:minCount 1 ; sh:nodeKind sh:IRI ] ;
    sh:property [ sh:path ex:locator ; sh:minCount 1 ] ;
    sh:property [ sh:path ex:supportedThrough ;
                  sh:maxCount 1 ; sh:datatype xsd:dateTime ] .
```

The inverse path finds assertions pointing to the claim; the second shape checks the assertion's source, locator and optional support horizon. These are deliberately local requirements, not universal rules for all RDF claims. The shape also

needs conditional constraints: `BoundedEnd` requires exactly one `validTo` and no `supportedThrough`; `UnboundedEnd` requires neither; and `UnknownEnd` requires no `validTo` and exactly one assertion horizon not before `validFrom`. Express those alternatives with `sh:xone`, or with a validation query joining each assertion to its claim. Check the end/start ordering and horizon/start ordering in that same validation layer. Targeting matters: if a malformed claim lacks the type that makes it a target, this shape alone may never inspect it. The data graph supplied to validation must also include the relevant assertions; validating an isolated claim graph could report missing evidence that exists elsewhere. This shape is illustrative and unexecuted.

Additional checks should test that an end follows the start under the chosen interval convention, source identifiers resolve to retained records, and claims expected to be exclusive do not overlap for the same scope. The last check depends on domain semantics and may require SPARQL-based constraints or a separate validation query. Shape conformance tells us only that selected conditions passed for the selected graph. A fabricated register row with a well-formed date can conform, and a correct but incomplete extraction can fail. Preserve the validation report and the shapes version with the release, and investigate failures rather than silently dropping records. See the SHACL Recommendation for targets, constraint components, and validation reports.

9.4 Describe Production History with the PROV Ontology (PROV-O)

Provenance records how a representation was produced and by whom. The PROV Ontology (PROV-O) describes entities, activities, and agents, with relations between them. In this model, a retained register version is a `prov:Entity`, extraction is a `prov:Activity`, and the software or responsible organisation may be a `prov:Agent`. The resulting assertion is another entity. A minimal extension to the earlier Turtle is:

```
ex:r1 a prov:Entity .
ex:extract-r1 a prov:Activity ;
    prov:used ex:r1 ;
    prov:wasAssociatedWith ex:extractor .
ex:extractor a prov:SoftwareAgent .
ex:assertion-r1-p1 prov:wasGeneratedBy ex:extract-r1 ;
    prov:wasDerivedFrom ex:r1 .
```

This describes a possible extraction history; the example does not pretend that the activity has actually run. In a real release, identify the source version, soft-

ware and configuration, run or review time, and responsible agent where known. If a person reviewed a candidate later, represent the review as a separate activity with its own inputs and outputs. The register's publisher, the person whose statement appears in minutes, the extractor, and the reviewer may all be different agents with different roles. PROV-O's qualified relations can attach roles or additional detail when a simple association is insufficient.

Provenance does not supply the authority rule. `prov:wasDerivedFrom` does not mean the register is correct, and `prov:wasAssociatedWith` does not identify an agent as the author of the original policy. Any authority rule must be recorded separately and scoped to the relevant question; the executable uses abstention on conflicting evidence. Nor does extraction time replace valid time: an assertion generated on 20 March can describe applicability on 1 March. If a register correction is ingested later, keep the earlier source and release identifiable so that both a current answer and a historical answer can be reconstructed.

9.5 Property Graphs and Cypher

A property graph presents nodes connected by typed, directed relationships. Nodes may have labels and properties; relationships may also have properties. In Neo4j, a relationship has one type and a direction, and Cypher can match it in a directed or undirected pattern. Direction is part of the model even when a particular query ignores it. The Cypher manual describes matching patterns and the results produced by Neo4j.

An initial projection could represent the register claim as `(v1:PolicyVersion)-[:APPLIES_TO {validFrom: date('2024-01-01'), validTo: date('2024-04-01'), endKind: 'bounded', polarity: true}]->(atlas:Project)`. This keeps the claim qualifiers near the relationship. For an `unknown` end, retain `endKind: 'unknown'` and put `supportedThrough` on the source-specific assertion; for an `unbounded` end, retain `endKind: 'unbounded'` and no horizon. It can work for a view whose source and release are known and whose edge identity and evidence are maintained. However, if each source has a separate edge, the graph must distinguish two sources making the same qualified claim from two different claims. If the relationship stores only one source field, multiple locators or extraction histories may be awkward to represent without duplication or loss. If another entity needs to refer to the assertion, a claim/assertion node design is often clearer.

For a closer match to the RDF example, use `(:ApplicabilityClaim)` nodes connected by `:POLICY_VERSION` and `:PROJECT`, with dates on the claim; `(:Assertion)-[:ASSERTS]->(:ApplicabilityClaim)` and `(:Assertion)-[:FROM_SOURCE]->(:Source)` preserve the separate assertion. The following illustrative query retrieves the same register-backed result. It assumes native date properties, a single source node with `id: 'R1'`, and a locator on each assertion:

```
MATCH (claim:ApplicabilityClaim)-[:POLICY_VERSION]->(version:PolicyVersion),
      (claim)-[:PROJECT]->(project:Project {id: 'atlas'}),
      (assertion:Assertion)-[:ASSERTS]->(claim),
      (assertion)-[:FROM_SOURCE]->(source:Source {id: 'R1'})
WITH claim, version, assertion, source, project, date('2024-03-01') AS day
WITH claim, version, assertion, source, project, day,
     CASE
       WHEN day < claim.validFrom THEN 'outside'
       WHEN claim.endKind = 'bounded' AND day < claim.validTo THEN 'applicable'
       WHEN claim.endKind = 'bounded' THEN 'outside'
       WHEN claim.endKind = 'unbounded' THEN 'applicable'
     WHEN claim.endKind = 'unknown' AND day <= assertion.supportedThrough THEN 'applicable'
       WHEN claim.endKind = 'unknown' THEN 'indeterminate'
       ELSE 'invalid'
     END AS temporalStatus
RETURN claim.id AS claimId, version.id AS versionId,
       assertion.id AS assertionId, source.id AS sourceId,
       assertion.locator AS locator, claim.validFrom AS start,
       claim.validTo AS end, claim.endKind AS endKind,
       assertion.supportedThrough AS supportedThrough,
       claim.polarity AS polarity, temporalStatus
ORDER BY versionId, assertionId
```

This query filters on the designated register and returns `indeterminate` for a validated unknown end after its horizon. Its `invalid` branch is not a substitute for checking required properties and date ordering: missing properties can evaluate as null and a recognised end kind can reach an earlier branch. Absence must not be mapped to `unbounded`. To show disagreement, run a separate query for the minutes and return its claim, temporal status and locator alongside the register's. To answer a two-hop path question, classify both premises and return the claim and assertion identifiers for *each* hop; the endpoints alone are insufficient evidence. Parameterise application inputs rather than constructing query strings from user-provided values. This Cypher is illustrative and unexecuted.

Cypher and SPARQL are query languages for different graph representations; Neo4j is a particular graph database product implementing Cypher. A property graph is not inherently Neo4j, and RDF is not inherently a particular vendor's triple store. Product versions and implementations differ in syntax, constraints, transaction guarantees, import options, security, and query planning. Select a

documented version for any reproducible example. Graph-shaped syntax makes traversals legible, but it does not guarantee that a query is fast or that a path is semantically meaningful.

9.6 What Is Lost When a Graph Is Simplified?

The following comparison assumes that the starting records contain the identified claims, assertions, and source information described above. It compares mappings, not the inherent expressive power of RDF, property graphs, or relational tables. Each richer representation can preserve the relevant information if the mapping is designed and tested to do so.

Mapping or projection	Retained	Lost or made ambiguous without extra fields
Qualified claim plus separate assertions	Version, project, interval, end kind, polarity, assertion IDs, source, locator and support horizon; disagreement can remain visible.	Nothing among these listed fields if all are mapped faithfully; extraction history, validity convention and release identity still need explicit representation.
One direct version-to-project edge with dates	Endpoints and stored interval, if the edge also carries end kind and polarity.	Source, locator, independent assertions, support horizon and indeterminate status unless separately attached; an unknown end may be mistaken for an unbounded one.
One unqualified version-to-project edge	Endpoints and relation type.	Interval, conditions, assertion identity, provenance, polarity, and whether the relation is merely reported or accepted.

Mapping or projection	Retained	Lost or made ambiguous without extra fields
Set of unique subject–predicate–object triples	Distinct triple values.	Repeated identical occurrences cannot encode the number or independence of sources; dates and evidence disappear unless modelled in additional triples.
Simple analytical graph collapsing edges by endpoints	Connectivity for the selected projection.	Parallel claims and their predicates, periods, polarity, sources, and sometimes direction, depending on the projection.
Name-only person node for the minutes' Alex	The surface name.	The difference between a mention and an identified person; merging with either directory person would invent an identity decision.

Loss is relative to the questions. A connectivity-only graph may be suitable for a labelled exploratory figure, provided the release states its selection rules and links back to the detailed records. It is unsuitable as the only evidence for who asserted a dated relationship. A round trip should compare stable identifiers, typed values, intervals, end kinds, support horizons, assertion multiplicity, source locators, polarity, unresolved mentions, and release information, not merely node and edge counts. If a target cannot represent a field directly, retain a documented sidecar or declare the export lossy; never silently reconstruct an unknown end as unbounded or otherwise missing information as fact.

9.7 Storage and Workload Determine the Boundary

The comparison above identifies information a mapping must preserve; it does not choose an engine. That choice also depends on which questions must be served, how often data changes, and the cost of operating and checking the resulting system.

A graph model describes data and query meaning; a storage engine chooses physical structures to retrieve it. A table can store nodes and edges as rows with identifiers and use joins to follow paths. An RDF store may index permutations of subject, predicate, object and graph to find triples for different binding patterns. A property-graph engine may use indexes to find starting nodes and adjacency-oriented structures to expand relationships. These are common design approaches, not universal guarantees: engines vary, and a query planner may combine indexes, scans, joins, caching and traversal operators. Neo4j documents multiple store formats, so do not infer current physical layout or performance from a generic description of an older release.

Starting-point selectivity matters. Finding Atlas by a stable indexed ID and expanding a small, bounded neighbourhood is a different workload from scanning all claims for a date range or aggregating across every policy. A high-degree node can make even a short traversal expensive; an unbounded path can produce a very large number of possible paths or encounter cycles. A relational query can be efficient for repeated fixed-depth joins, and a graph database can also scan and aggregate. Measure actual plans, cardinalities, cache conditions, ingestion costs, and result size rather than assuming that one physical model wins every query.

DuckDB's role in the Northbridge construction route is to hold and query the tabular records produced by Python: it supports joins, filtering and aggregation, and its SQL also supports recursive common table expressions for iterative queries. It is not necessary to add a server merely because the data represents a graph. A separate RDF serving layer becomes attractive when shared RDF identifiers, a SPARQL interface, RDF interchange, or SHACL validation are central requirements. A property-graph service becomes worth testing when interactive neighbourhood and path queries, graph-oriented application development, or a particular operational requirement justify it. Neo4j is one concrete choice within the latter category, not a synonym for a graph database.

Compare candidates using the *same* versioned data and questions. Include a register-scoped date query, a two-hop path returning evidence for every edge, a minutes disagreement query, and an unresolved-mention query. Add representative larger synthetic distributions only if they preserve useful properties such as degree skew and assertion multiplicity. Measure build and update time, storage, warm and cold query latency, throughput under expected concurrency, and the cost of exporting evidence. Fix hardware, versions, indexes, dataset, and result semantics; inspect plans and report failures as well as successful timings. A tiny Northbridge example tests correctness and integration, not production scalability. Include backup, access control, deployment, and skills in the decision, because a modest query-time improvement can be outweighed by the cost of maintaining another service.

9.8 A Bounded Future Neo4j Smoke Test

A later integration can test the mapping without making Neo4j a prerequisite for the canonical offline route. Use an isolated, single-node Neo4j Community container with an explicitly pinned image version or digest and a disposable database volume. Bind the service only as needed for the local test, use test-only credentials supplied through the test environment, and remove the test container and its disposable data after the run. Record the image, driver and Cypher versions and exact command in the repository's container workflow. This is a proposed test design, not an executed test or a production configuration.

Load the same frozen fixture through a documented mapping using stable application IDs for versions, projects, claims, assertions and sources. Preserve the original source locators, date values, polarity, and unresolved mentions; do not use internal database IDs as external identity. Create a uniqueness constraint on the selected stable IDs where supported by the pinned edition, and explicitly test required fields in the loader or validation query. A uniqueness constraint alone does not require every node to carry the property. Make loading idempotent under the chosen keys, and verify that a second load creates no extra nodes, relationships, or assertions. Consult the pinned version's constraint documentation for exact syntax and availability.

The smoke test has bounded acceptance cases:

1. Compare the loaded claim and assertion IDs, qualifiers, polarity, source IDs and locators with the frozen canonical export; fail on missing or extra

records rather than merely comparing totals.

2. Query a dated applicability claim and a fixed two-hop path, returning the evidence for each hop. Compare ordered results with the canonical implementation under the *same* selection and conflict rules.
3. Return the minutes' competing assertion and its locator without deleting or silently resolving it; retain the fixture's unresolved Alex mention.
4. Reject or report an unknown endpoint, malformed interval, missing evidence, and duplicate stable ID. Test an end boundary on 1 April explicitly.
5. Apply a small, explicitly defined correction as a new source or assertion version, verify the changed current answer, and verify that the earlier release remains reconstructible. If release history is not yet implemented, mark this case blocked rather than claiming success.

Exercise only the fixed query patterns needed for these cases, using parameterised inputs and a timeout. Capture exit status and assertion failures, and clean up the test's own resources even on failure. This verifies mapping and basic operation on the fixture; it does not establish availability, security, disaster recovery, production throughput, or a performance advantage. Neither this design nor the illustrative Cypher above is evidence that Neo4j has been run.

9.9 Analyse a Projection in R

The companion `examples/r/knowledge_graph.R` reads the eight entities and five precomputed positive-projection relations from `examples/fixture.json` using `jsonlite`. It uses `igraph` to create a directed graph and `tidygraph` to calculate node summaries. This removes the earlier hard-coded copy of those rows, but does not make R consume the Python source-derived rebuild: a changed `SOURCE_RECORDS_PATH` can change `c1` in DuckDB while the R script still reads the unchanged fixture. The projection must be regenerated or compared with the rebuilt claims before treating the analyses as synchronised. It intentionally drops end kind, support horizon, polarity, provenance and claim identity; it is therefore undated and cannot implement the indeterminate answer policy. See the official igraph data-frame conversion reference and tidygraph activation reference for the operations used here.

Run the companion from the repository root with `just examples knowledge_graph knowledge-graph-r`. The command executes the pinned R runtime

and checks the path, component count, vertex count, and edge count. For an interactive exploration, open the script after the check and inspect the printed `node_summary`; the checked command is the reproducible baseline, while interactive filtering and plotting are exploratory operations.

The essential construction is short because the graph has already been reduced to an edge list, not because the modelling problem has disappeared. The fixture's `entities` become the vertex table and preserve the two isolated Alex entities; its `relations` preserve edge types and unit weights for this particular structural analysis. `graph_from_data_frame()` matches edge endpoints to the vertex table's `name` column and creates a directed graph; it does not infer that two equal labels denote the same person.

R:

```
graph <- graph_from_data_frame(
  relations,
  directed = TRUE,
  vertices = entities |> rename(name = id)
)

path <- shortest_paths(
  graph,
  from = "policy:travel-v1",
  to = "org:northbridge",
  weights = E(graph)$weight,
  mode = "out",
  output = "vpath"
)$vpath[[1]]

weak_components <- components(graph, mode = "weak")
```

The companion converts the path vertices to labels and reports `Northbridge Travel Policy v1 -> Atlas -> Northbridge`, with unit cost 2. It reports three weak components: the main group and the two isolated Alex vertices. The `tidygraph` transformation then adds total degree and component membership to each node, which is useful for inspection but does not change the evidence represented by the graph.

This is a projection analysis, not a second claim store. The path is a sequence of edges in the selected positive topology; it does not establish that the policy was valid on the requested date, that the register outranks the minutes for every purpose, or that the path is causal. The example therefore demonstrates both the value of graph operations and the reason to return to the DuckDB claims and assertions when an answer needs dates, polarity, provenance, or source-specific evidence.

Build the positive projection.

The edge list has one row per directed relation. Its five rows appear in the edge-list table; each has weight 1. The vertex table also supplies all eight identifiers, including the two Alex entities that have no edges. Passing that table to `graph_from_data_frame()` preserves those isolated vertices rather than constructing vertices solely from edge endpoints.

Source	Relation	Target
`policy:travel-v1`	`applies_to`	`project:atlas`
`policy:travel-v1`	`owned_by`	`person:morgan-lee`
`project:atlas`	`belongs_to`	`org:northbridge`
`policy:travel-v2`	`applies_to`	`project:atlas`
`project:orion`	`belongs_to`	`org:northbridge`

Interpret paths and components.

Starting at `policy:travel-v1`, the outgoing `applies_to` edge reaches Atlas; its outgoing `belongs_to` edge reaches Northbridge. The two edge weights add to 2. The `owned_by` edge leads to Morgan Lee and does not continue to Northbridge in this projection. Reversing the question is different: `mode = "out"` cannot traverse these edges backwards. For weak components, however, ignore direction while grouping vertices connected by any sequence of edges. The first six entities form one component, and each isolated Alex entity forms a component of one, giving three in total. The node summary's total degree counts incident edges regardless of direction; it is a structural count, not a measure of a person's importance or the quality of the evidence.

The script asserts a directed shortest path from version 1 through Atlas to Northbridge, with total cost 2 under its unit weights. It also asserts three weakly connected components: the main connected group and the two isolated Alex entities. Weak connectivity ignores edge direction; the shortest path uses outgoing edges. Neither calculation establishes the authority, temporal validity or truth of the path. Weights here are assigned unit costs, not confidence scores.

The projection collapses the positive `c4` and `c6` assertions into one version-2 edge, discards `c5` and all dates, source IDs and locators, and retains `c1` even outside its validity interval. The source correction that moves `c1`'s end to 1

March changes the DuckDB answer on 15 March but leaves this R topology unchanged. That difference is expected for an undated projection, and the R run does not verify the corrected source input. The R graph contains no claim IDs on edges, so a path cannot itself trace each step back to its evidence; return to the claims table or build an evidence-preserving mapping. A dated or source-selected analysis requires a different, explicitly defined projection.

10 Validate, Evaluate, and Diagnose

This section separates three questions that are often conflated: whether the records satisfy declared constraints, whether rules derive a conclusion, and whether a source supports an assertion. It then turns those distinctions into evaluation cases and release decisions, so a passing database check cannot be mistaken for evidence that the graph is true or complete.

10.1 Validation, Inference and Verification

These operations answer different questions. **Validation** asks whether records satisfy declared constraints. The executable rejects duplicate IDs, unknown source and entity references, invalid date and end-kind combinations, invalid polarity or empty predicates, and a claim mention that fails the source-and-participant check. It also checks that the `relations` entries are unique, have positive weights and equal the set of positive claim triples. Its negative tests mutate copies of the fixture to exercise unknown entities, malformed temporal fields, a mismatched mention, a duplicate ID and a selected candidate on an abstention. The construction slice separately checks exact source spans, accepted/rejected/abstained identity decisions and a refused unsupported date phrase. Passing those checks says that these particular rules held for this input; it does not establish that the source text actually supports any claim.

The refusal and validation trace makes the boundary observable. The retained sentence at `S2:1`, “Travel Policy v1 applies to Atlas from sometime next spring.”, matches none of the supported date and sentence patterns. The executable returns no candidate for that record and a refusal with its source, method and reason `no supported date and sentence pattern`; it does not invent a date. The diagnostic fixture comparison reports a changed constructed claim without preventing a source-derived build, while the ledger validator checks its own

structural rules, including the fixture's references and temporal combinations; mutation tests confirm that specific invalid copies are rejected. This is a sequence of executable checks, not a human review of source truth. The refusal is separate from the seven curated-and-constructed ledger rows.

The projection check compares positive triples, not dates, provenance, credibility or the numerical meaning of weights. As the R analysis illustrates, passing it does not make the projection a source-resolved, dated view.

The same distinction helps diagnose a failed answer. If the minutes' negative assertion `c5` is absent, inspect retrieval and extraction before changing the graph schema. If `c5` is present but the answer ignores it, inspect the evidence-selection or conflict policy. If the answer cites `c1` for a date after its bounded end, inspect temporal filtering. If Morgan is linked correctly but Alex is merged without an identifier, inspect identity resolution and its abstention rule. Each failure points to a different repair and should be recorded with the source span, intermediate claim or candidate decision, selected evidence, and policy version that produced the answer.

Use the observed answer to choose the next inspection. The following decision table makes the diagnostic route concrete:

Observed answer on 15 March	Inspect first	Evidence to retain	Targeted repair and retest
`supported`, with only `c1` selected	Candidate extraction and negative-pattern handling	Retained `M1:paragraph-3` span, candidate list, extraction method and policy revision	Restore the negative `c5` candidate, rebuild, and rerun the conflict query
`abstain: conflicting evidence`, but no `c5` locator	Evidence joins and source/claim linkage	Claim IDs, source IDs, locators and the query result before aggregation	Repair the evidence link or retrieval boundary, then rerun the same question

Observed answer on 15 March	Inspect first	Evidence to retain	Targeted repair and retest
`supported` for a date after `c1` ends	Temporal predicate and end-kind validation	`valid_from`, exclusive `valid_to`, requested date and temporal classification	Correct the date filter or interval data, then test the boundary dates again
Morgan returned as the 1 March owner without qualification	Unknown-horizon handling and answer policy	`c2` support horizon, requested date and selected answer rule	Preserve the recorded 1 January owner but return indeterminate continuation

The table separates a missing assertion from a present assertion that is misinterpreted. Changing the schema cannot recover a candidate that extraction discarded, and adding another edge cannot justify a date that the evidence does not support. A retest should use the same question, source boundary and policy version so that the change in answer can be attributed to the repair.

Inference derives a conclusion from premises under an explicit rule. For example, a system could define a rule that turns an `applies_to`/`belongs_to` path into a carefully named organisational association. The current Python example only returns the path; it does not implement that rule or materialise a new claim. Before adding it, specify its meaning, acceptable predicates, scope, source restrictions and temporal overlap. If two half-open input intervals overlap, the derived relationship can hold at most during their intersection; if they do not overlap, the path is not evidence of a simultaneous relationship. Record both premise IDs, the rule version and the derivation run, so a corrected premise or rule can invalidate the result. A derived association is not the same as an explicit statement in the register.

Verification checks whether an assertion is supported by the appropriate evidence and is adequate for the question. A reviewer could locate the retained register version, inspect its Atlas entry, confirm the version and interval, and record the review and its limitations. Checking that `c1.source` refers to `s-`

`register` is validation; comparing `c1` with the actual register is verification. The synthetic excerpts permit comparison with the retained sample text, but neither they nor the fixture supply complete underlying documents or independent external evidence, so they cannot demonstrate that external verification step. It can demonstrate the structure of a review workflow and the need to abstain when the available evidence conflicts.

For RDF data, SHACL specifies validation of a data graph against shapes and produces a validation report. It does not make a conforming graph factually correct. Likewise, the DuckDB schema above and the Python checks are not a SHACL implementation. If a system validates inferred data, record whether validation ran on asserted data only or on an expanded graph, and keep the underlying asserted data distinguishable from derived results.

Gold sets and useful measures.

Build an evaluation set from questions and independently adjudicated evidence, not merely from rows copied from the graph being tested. A useful gold item contains the question, entity and predicate scope, requested valid date, source boundary, expected answer or abstention, supporting and opposing evidence identifiers, and an explanation of the adjudication. Record who reviewed it, the evidence version and the policy version. The fixture's date cases are useful regression examples, but their expected outputs were chosen from the same synthetic input and policy; passing them does not estimate accuracy on unfamiliar documents.

10.2 Work one adjudication before scoring

Consider Q4, agreement on version 1 for Atlas on 15 March. An independent adjudicator would first freeze the question, source boundary, and conservative answer policy without seeing a system result. They would then inspect the retained evidence and write a gold item such as:

Gold-item field	Q4 adjudication
Question and scope	Does the register and the minutes agree that version 1 applies to Atlas on 15 March? Use both named sources and the requested date.

Gold-item field	Q4 adjudication
Supporting/opposing evidence	Positive `c1` from `s-register`, `R1:P1-P2`; negative `c5` from `s-minutes`, `M1:paragraph-3`.
Temporal decision	Both assertions are applicable on 15 March under the declared half-open and unknown-horizon rules.
Gold answer	`abstain: conflicting evidence`; retain both claim IDs and locators.
Adjudication record	Reviewer, evidence revision, policy revision, reasoning, and any unresolved disagreement.

Only after that record is frozen should the system answer the same question. Compare the complete unit: answer label, selected evidence, and required abstention. A system that returns `abstain: conflicting evidence` but cites only `c1` has an evidence-retrieval or citation failure even though its label looks correct. A system that returns `supported` has a policy or aggregation failure because it ignored the opposing applicable assertion. A system that returns `abstain: no evidence` has a retrieval or temporal-filter failure because evidence was available. This is the adjudication operation the scores are meant to evaluate.

The unit becomes clearer when one adjudicated item is compared with a system answer rather than only with a final score. Suppose an independent reviewer has frozen Q4 as `abstain: conflicting evidence`, with both `c1` and `c5` and their register and minutes locators. These outcomes are different evaluation results:

System answer	Judgement	Failure diagnosed
`abstain: conflicting evidence`, citing `c1` and `c5`	Match	The label, selected evidence and required abstention agree with the gold item.

System answer	Judgement	Failure diagnosed
`abstain: conflicting evidence`, citing only `c1`	Mismatch	The answer label is plausible, but retrieval or evidence selection omitted the opposing assertion.
`supported`, citing only `c1`	Mismatch	The answer policy ignored an applicable negative assertion; this is not repaired by a correct-looking citation.
`abstain: no evidence`	Mismatch	Retrieval or temporal filtering failed because the adjudicated evidence was available.

The comparison shows why evaluation must score the complete answer unit. A label-only score would count the first three rows too generously and would not tell an author whether to inspect extraction, evidence joins, or answer policy. The current Q1–Q5 table below remains a synthetic teaching walkthrough; it is not this independent adjudication exercise.

The current Q1–Q5 table cannot claim this independent procedure was performed: its answers and adjudications were constructed from the same fixture and policy for teaching. It is therefore a worked scoring walkthrough and regression expectation, not a gold-set accuracy result. To turn it into an evaluation, freeze labels from retained source evidence before running the system, include cases from a held-out source family, and retain disagreements rather than silently converting them into a consensus label.

For the conservative target policy, a small regression table is:

Question for version 1 and Atlas	Expected selected evidence	Expected policy output
31 December 2023	none	`abstain: no evidence`

Question for version 1 and Atlas	Expected selected evidence	Expected policy output
15 March 2024	`c1` positive, `c5` negative; both applicable	`abstain: conflicting evidence`
16 March 2024	`c1` applicable, `c5` indeterminate	`abstain: indeterminate temporal scope`
1 April 2024	`c1` outside, `c5` indeterminate	`abstain: indeterminate temporal scope`
15 April 2024	`c1` outside, `c5` indeterminate	`abstain: indeterminate temporal scope`

Add a boundary case for 1 April: `c1` is excluded at its bounded end, while the unknown-ended `c5` is indeterminate after its support horizon. A dated two-hop case should classify both edge intervals; an outside premise makes the path outside, while an indeterminate premise makes the path indeterminate. A provenance case should check source, locator and mention, and that a missing mention such as `c7` does not eliminate the claim. The fixture records `m-alex` as unresolved and `r2` as abstaining between two candidates; the validator checks this representation, while the separate bounded resolver executes the synthetic identity cases described above.

Separate stages in the scorecard. For extraction, compare proposed source assertions with independently reviewed assertions and count correct, missing and spurious records. For entity resolution, evaluate links and non-links, especially same-name people and changing names. For retrieval, assess whether all required supporting and opposing passages were returned. For final answers, measure correctness with respect to the specified evidence and policy, as well as whether the cited evidence actually supports each answer. A correct label with an unrelated citation is an evidence failure. Report results by slice, including ambiguous identities, negative claims, conflicts and temporal boundaries; aggregate accuracy can conceal the cases that motivate a knowledge graph.

Precision is correct returned items divided by returned items; recall is correct returned items divided by all gold items that should have been returned. Define the matching unit before calculating either: a claim ID, a source-qualified assertion, a date-qualified answer or an evidence span can lead to different counts. A system can improve apparent precision by answering less often, so report an-

swer coverage and the correctness of answered cases alongside the number and reasons for abstentions. If the denominator is zero, report the count and mark the ratio undefined rather than inventing a value. For a small gold set, publish counts rather than implying that a percentage gives a precise population estimate.

The following synthetic answer set applies the opening Q1–Q5 contracts to the retained Northbridge evidence. The adjudications and system answers are constructed for teaching from the same fixture and policy, not independently labelled or collected from five end-to-end system runs. The check status distinguishes executable evidence from illustrative answer presentation.

Question scope	Gold/adjudicated answer and evidence	Illustrative system answer	Abstention assessment and check status
Q1: register version for Atlas on 1 March	Version 1, `c1`, `s-register`, `R1:P1-P2`; its bounded interval contains the date.	Version 1, with `c1` and the register locator.	No abstention required. The `c1` date classification is asserted; presentation is illustrative.
Q2: register owner of version 1, including continuation on 1 March	Morgan Lee on 1 January, `c2`, `s-register`, `R1:P1-P2`; continued ownership on 1 March is indeterminate because the unknown end is supported only through 1 January.	Morgan Lee on 1 January; abstain on continued ownership on 1 March, with `c2` and the register locator.	Appropriate partial abstention on continuation. The ledger row is validated; the owner answer is illustrative.

Question scope	Gold/adjudicated answer and evidence	Illustrative system answer	Abstention assessment and check status
Q3: what the minutes state on 15 March	Version 2 positive (`c4`) and version 1 negative (`c5`), `s-minutes`, `M1:paragraph-3`, each with a one-day support horizon.	Both statements, their polarity, date, support horizons and minutes locator.	No abstention required for reporting the statements. Source-derived rows are asserted; the combined answer is illustrative.
Q4: agreement on version 1 for Atlas on 15 March	`c1` positive (`R1:P1-P2`) and `c5` negative (`M1:paragraph-3`), both applicable; abstain on the disputed applicability under the conservative policy.	`abstain: conflicting evidence`, with `c1`, `c5` and both locators.	Appropriate abstention on applicability; the dated conflict answer is asserted by the executable.
Q5: identity of the Alex Kim attendee	Unresolved `m-alex`, `r2`, both candidate person IDs and `M1:paragraph-3`; assign neither person to attendance.	Abstain on identity, returning the unresolved mention, decision, candidates and locator.	Appropriate abstention on identity; the bounded resolver is executable, while disclosure is illustrative.

Score one complete question as one unit: a match requires the stated answer, evidence and any required abstention. On these constructed answers, illustrative **full-answer coverage** is 2/5: Q1 and Q3 answer every requested part without abstention; Q2 is partial, and Q4 and Q5 abstain on the disputed de-

cision. The other three remain in the denominator. Illustrative **policy-match accuracy**, counting an appropriate abstention as correct and an unnecessary or missing abstention as incorrect, is 5/5: all five constructed answers match their adjudications, including the partial abstention in Q2. These counts describe this designed example only; they do not estimate population accuracy or replace independent evaluation.

The date rows above and the separate 31 December, 15 March, 16 March and 1 April cases are **regression checks** against the synthetic fixture and a specified policy. Their pass count measures consistency with those chosen expectations. It does not measure extraction accuracy, population answer precision or recall. To measure those, freeze a question unit and source boundary, obtain independent labels from retained evidence, run the system without seeing those labels, then count correct returned answers, unsupported returned answers, missed answerable items and appropriate abstentions. The deliberately incomplete 15 March run below demonstrates one missed negative claim and a changed answer, but one designed failure is not an accuracy estimate.

Construct a held-out set from source records or document families not used to tune extraction and matching rules. Splitting individual rows from the same document across development and test can leak the same wording, entities and corrections into both. Review difficult cases independently, resolve disagreements using written criteria, and retain unresolved or disputed labels rather than forcing a false consensus. Include negative examples and missing-evidence cases, but label absence as negative only when the gold-set scope justifies that conclusion. Re-run the same gold set after changing the schema, extraction pipeline, linking rules, answer policy or sources; investigate regressions before replacing expected answers.

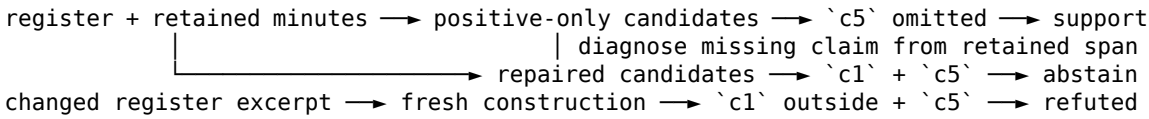
11 Maintain, Evaluate, Release, and Govern

A release is a reproducible combination of input versions, transformations, schema and answer policy, not merely a database file. For the fixture, the inputs include `examples/source_records.json`, `examples/fixture.json`, `examples/python/construction.py`, `examples/python/knowledge_graph.py`, the manifest and the primer-specific runtime definition. Record their revisions or digests, the runtime and dependency versions, validation results, evaluation

results and the exact build command. The current example uses fresh temporary DuckDB files and tests that they can be reopened; it does not create a release manifest, preserve input snapshots after the run or publish a database. Those are production extensions, not features to infer from the example.

The executable builds three *temporary* databases for the same 15 March version 1 question. Its first run deliberately filters out negative candidates despite retaining the minutes text; its second restores `c5`; its third substitutes a synthetic register sentence with an earlier exclusive end date and rebuilds from that changed source input. These are asserted before-and-after results, not published or durable releases. The source substitution is a hypothetical correction for teaching; the script does not establish that Northbridge issued one.

Temporary run	Selected evidence on 15 March	Asserted answer	Diagnosis
Deliberately incomplete extraction	`c1` positive and applicable; `c5` absent from the loaded graph despite its retained `M1:paragraph-3` span.	`supported`	Positive-only candidate selection omitted the negative sentence.
Repaired extraction	`c1` positive and `c5` negative, both applicable; original `R1:P1-P2` and `M1:paragraph-3` inputs.	`abstain: conflicting evidence`	Rebuilding with negative candidates restores the opposing assertion.
Hypothetical source correction	`c1` outside after its new 1 March exclusive end; `c5` negative and applicable.	`refuted`	A changed register excerpt produces a different source-derived `c1` in a fresh database.



The diagnosis is explicit: the minutes' negative sentence is retained and its exact span is checked, but the deliberately incomplete candidate set drops `c5`. The repaired rebuild includes `c5` and reruns the question. The separate source substitution changes only the register sentence ending `2024-04-01` to one ending `2024-03-01`; the script asserts the rebuilt `c1` end and the `refuted` answer. The changes arise at different stages, and the run prints the selected claim IDs and classifications. The three databases are discarded when the temporary directory closes.

For a real release, the important operation is to record the new source and assertion version, identify the release that adopted it, and trace affected answers, indexes, projections and cached results. The example keeps separate databases only during one run; it does not implement durable release identities, recorded times, an approval step or dependency invalidation. If a downstream path relied on `c1`, a release process would need to recompute it or mark it stale. A checksum or current database snapshot alone cannot reconstruct an earlier answer; the permitted source versions and transformation policy must remain available.

11.1 Diagnose, correct, and apply a release

Use a failure to locate the stage that needs attention. A missing source passage may call for retrieval or source-coverage work; a retrieved passage with a missing claim points towards extraction; a correct claim attached to the wrong person points towards identity resolution. If the evidence is present but an answer ignores a conflicting assertion or an uncertain interval, inspect selection and answer policy before changing the extractor. Conversely, a valid-looking citation that does not support the answer is an evidence-linking failure even when the answer happens to match the gold label. Preserve the question, selected evidence, intermediate decisions and policy version so that these possibilities can be distinguished rather than treating every wrong answer as a graph-storage problem.

For an operational graph, ingest a new source version into a staging area; validate identifiers, references and intervals; compare the candidate with the previous release; run the evaluation set; and review material changes before public-

ation. The diff should show added, changed and withdrawn assertions, affected entities and derived paths, changes in source coverage and the effect on representative questions. Build into a new versioned location rather than overwriting the only copy. Publish the version and its provenance together, and retain a tested route back to the prior release subject to retention and access rules. A reproducible build requires access to permitted input versions as well as code; a digest can identify missing material but cannot reconstruct it.

Corrections, retractions, and dependencies.

A correction, an end of validity and a retraction are different events. Ending `c1` at 1 April describes the asserted policy interval; it does not say that the register was wrong when it made the assertion. A later correction might revise that interval and require a new source version. A retraction says that an assertion or source should no longer be relied upon under a stated policy. Preserve a record of the change, its effective and recorded times, reason, actor and replacement when applicable. Whether the old content itself can be retained depends on legal, privacy and retention obligations; a tombstone or restricted audit record may be needed where the original must be removed. Do not silently delete an assertion and leave derived results or cached answers looking unchanged.

Trace dependencies from a changed assertion to derived claims, projections, indexes and published answers. On retraction, recompute or invalidate those dependants and reassess answers that relied on them. A derived result with two independent supporting paths may survive one withdrawn premise, but only if the remaining path still satisfies the rule and evidence policy. This dependency tracking is a proposed production feature; the Python fixture does not maintain it. For mutable sources, keep the old source version and locator where permitted so that a reviewer can explain what a previous release knew, even if the preferred answer has since changed.

11.2 Responsibility, Access, and Boundaries

Assign named responsibility for source admission, identity resolution, schema and predicate definitions, conflict policy, release approval and incident response. A data steward need not perform every task, but decisions require an accountable owner and a documented escalation route. Restrict access to sensitive sources and evidence spans, avoid publishing private attributes merely because they can be joined, and check whether downstream users can infer sensitive

information from paths. Govern identifiers, corrections and deletion across replicas and caches, not only in the primary database. Review retention, permissions and intended use with the relevant owners before sharing a release.

Finally, communicate the graph's boundary with each release: which sources and dates it covers, which claims are asserted or derived, how conflicting evidence is presented, which validation and evaluation checks passed, and where a reader can report a suspected error. A reliable graph is one whose answers can be challenged and repaired; the presence of a graph database or a successful schema check is only one part of that process.

For this milestone, the omitted pieces are intentional: Neo4j execution remains a future pinned-container smoke test; the current Python example implements a bounded correction and rebuild but not a full source extractor, durable release manager or retraction engine; the R companion reads a precomputed positive projection from the checked-in fixture and needs an explicit synchronisation check after source-derived changes; and the synthetic fixture cannot support a population coverage or accuracy study. The construction diagram and comparison tables carry the explanatory burden in this text. These are scope dispositions, not claims that the omitted capabilities are complete.

12 Further Reading

- RDF 1.1 Concepts and Abstract Syntax defines RDF graphs, triples, identifiers, literals, and the distinction between syntax and model.
- Defining N-ary Relations on the Semantic Web explains patterns for relations that need more than two participants or qualifiers.
- PROV-O provides vocabulary for describing provenance across entities, activities, and agents.
- SPARQL 1.1 Query Language and SHACL specify RDF graph querying and shape-based validation, respectively.
- Estimating the False Negatives Due to Blocking in Record Linkage examines candidate-selection error, a distinct source of identity-resolution failure.
- DuckDB documentation and the Neo4j Cypher manual provide implementation details that must be checked against the versions actually pinned for examples.